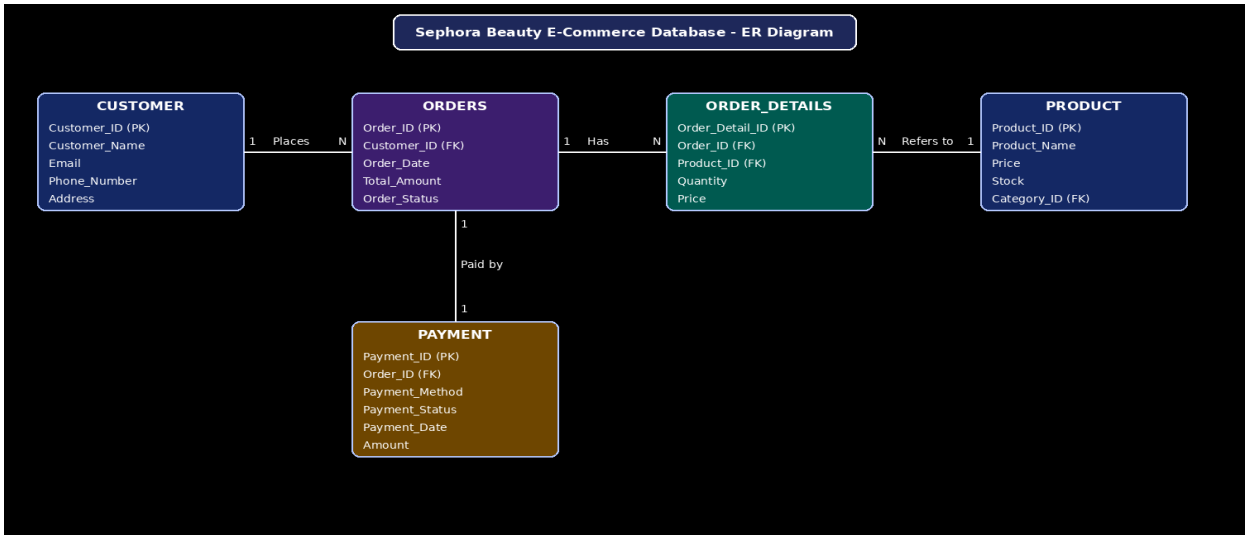
Sephora Beauty E-Commerce Database Management System

DATABASE RELATIONSHIP ANALYSIS USING JOINS

CHAPTER-9

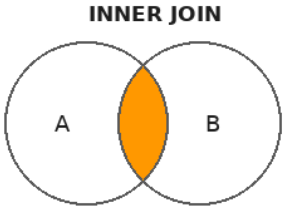
1.Combining Multiple Tables:

The Sephora Beauty database contains different tables that store customer, product, order, and payment information. The tables can be combined using JOIN operations to retrieve related information.



2. INNER JOIN:

INNER JOIN returns only the records that have matching values in both tables.



2.1 Customer and Orders:

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.CUSTOMER_NAME,
  3  O.ORDER_ID,
  4  O.ORDER_DATE,
  5  O.TOTAL_AMOUNT,
  6  O.ORDER_STATUS
  7  FROM CUSTOMER C
  8  INNER JOIN ORDERS O
  9  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  ORDER BY O.ORDER_ID;
```

```
CUSTOMER_ID CUSTOMER_NAME ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- ------------- -------- --------- ------------ ------------
        101 Priya              201 10-SEP-26         1797 Delivered

        102 Ananya             202 11-SEP-26         1299 Shipped

        103 Kavya              203 12-SEP-26          799 Delivered

        101 Priya              204 23-SEP-26         1299 Shipped

        102 Ananya             205 25-SEP-26          899 Placed
```

2.2 Orders and Payment:

```
SQL> SELECT O.ORDER_ID,
  2  O.CUSTOMER_ID,
  3  O.ORDER_DATE,
  4  O.TOTAL_AMOUNT,
  5  P.PAYMENT_ID,
  6  P.PAYMENT_METHOD,
  7  P.PAYMENT_STATUS,
  8  P.AMOUNT
  9  FROM ORDERS O
 10  INNER JOIN PAYMENT P
 11  ON O.ORDER_ID = P.ORDER_ID
 12  ORDER BY O.ORDER_ID;
```

```
ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_ID PAYMENT_METHOD   PAYMENT_STATUS AMOUNT
-------- ----------- --------- ------------ ---------- ---------------- -------------- ------
     201         101 10-SEP-26         1797        401 UPI              Success          1797

     202         102 11-SEP-26         1299        402 Credit Card      Success          1299

     203         103 12-SEP-26          799        403 Debit Card       Success           799

     204         101 23-SEP-26         1299        404 Cash on Delivery Pending          1299
```

2.3 Product and Order_Details:

```
SQL> SELECT P.PRODUCT_ID,
  2  P.PRODUCT_NAME,
  3  P.PRICE,
  4  OD.ORDER_ID,
  5  OD.QUANTITY,
  6  OD.PRICE AS UNIT_PRICE
  7  FROM PRODUCT P
  8  INNER JOIN ORDER_DETAILS OD
  9  ON P.PRODUCT_ID = OD.PRODUCT_ID
 10  ORDER BY OD.ORDER_ID;
```

```
PRODUCT_ID PRODUCT_NAME          PRICE ORDER_ID QUANTITY UNIT_PRICE
---------- -------------------- ----- -------- -------- ----------
       501 Matte Lipstick         499      201        2        499

       502 Volume Mascara         799      201        1        799

       503 Vitamin C Serum       1299      202        1       1299

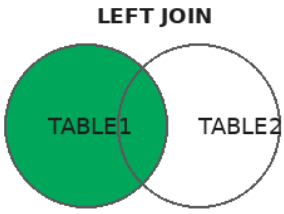
       502 Volume Mascara         799      203        1        799

       503 Vitamin C Serum       1299      204        1       1299

       504 Hydrating Moisturizer  899      205        1        899
```

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.



3.1 Display All Customers and Their Orders:

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.CUSTOMER_NAME,
  3  O.ORDER_ID,
  4  O.ORDER_DATE,
  5  O.TOTAL_AMOUNT,
  6  O.ORDER_STATUS
  7  FROM CUSTOMER C
  8  LEFT JOIN ORDERS O
  9  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  ORDER BY C.CUSTOMER_ID;
```

```
CUSTOMER_ID CUSTOMER_NAME ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- ------------- -------- --------- ------------ ------------
        101 Priya              201 10-SEP-26         1797 Delivered

        101 Priya              204 23-SEP-26         1299 Shipped

        102 Ananya             202 11-SEP-26         1299 Shipped

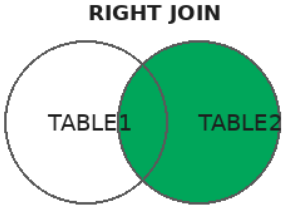
        102 Ananya             205 25-SEP-26          899 Placed

        103 Kavya              203 12-SEP-26          799 Delivered

        104 Meera
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.



4.1 Display All Orders and Customer Details:

```
SQL> SELECT O.ORDER_ID,
  2  O.ORDER_DATE,
  3  O.TOTAL_AMOUNT,
  4  O.ORDER_STATUS,
  5  C.CUSTOMER_ID,
  6  C.CUSTOMER_NAME,
  7  C.EMAIL
  8  FROM CUSTOMER C
  9  RIGHT JOIN ORDERS O
 10  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11  ORDER BY O.ORDER_ID;
```

```
ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS CUSTOMER_ID CUSTOMER_NAME EMAIL
-------- --------- ------------ ------------ ----------- ------------- ---------------
     201 10-SEP-26         1797 Delivered            101 Priya         priya@gmail.com

     202 11-SEP-26         1299 Shipped              102 Ananya        ananya@gmail.com

     203 12-SEP-26          799 Delivered            103 Kavya         kavya@gmail.com

     204 23-SEP-26         1299 Shipped              101 Priya         priya@gmail.com

     205 25-SEP-26          899 Placed               102 Ananya        ananya@gmail.com
```

5. Complete Order Details :

This query combines Customer, Orders, Order_Details, Product, and Payment tables.

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.CUSTOMER_NAME,
  3  O.ORDER_ID,
  4  O.ORDER_DATE,
  5  P.PRODUCT_NAME,
  6  OD.QUANTITY,
  7  OD.PRICE AS UNIT_PRICE,
  8  OD.QUANTITY * OD.PRICE AS ITEM_TOTAL,
  9  O.TOTAL_AMOUNT,
 10  PY.PAYMENT_METHOD,
 11  PY.PAYMENT_STATUS
 12  FROM CUSTOMER C
 13  INNER JOIN ORDERS O
 14  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 15  INNER JOIN ORDER_DETAILS OD
 16  ON O.ORDER_ID = OD.ORDER_ID
 17  INNER JOIN PRODUCT P
 18  ON OD.PRODUCT_ID = P.PRODUCT_ID
 19  LEFT JOIN PAYMENT PY
 20  ON O.ORDER_ID = PY.ORDER_ID
 21  ORDER BY C.CUSTOMER_ID, O.ORDER_DATE DESC;
```

```
CUSTOMER_ID CUSTOMER_NAME ORDER_ID ORDER_DAT PRODUCT_NAME          QUANTITY UNIT_PRICE ITEM_TOTAL
----------- ------------- -------- --------- -------------------- -------- ---------- ----------
TOTAL_AMOUNT PAYMENT_METHOD   PAYMENT_STATUS
------------ ---------------- --------------
        101 Priya             204 23-SEP-26 Vitamin C Serum             1       1299       1299
        1299 Cash on Delivery Pending

        101 Priya             201 10-SEP-26 Matte Lipstick              2        499        998
        1797 UPI              Success

        101 Priya             201 10-SEP-26 Volume Mascara              1        799        799
        1797 UPI              Success

        102 Ananya            205 25-SEP-26 Hydrating Moisturizer        1        899        899
         899

        102 Ananya            202 11-SEP-26 Vitamin C Serum             1       1299       1299
        1299 Credit Card      Success

        103 Kavya             203 12-SEP-26 Volume Mascara              1        799        799
         799 Debit Card       Success
```

6. Customer Purchase History:

This query displays the products purchased by each customer.

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.CUSTOMER_NAME,
  3  O.ORDER_ID,
  4  O.ORDER_DATE,
  5  P.PRODUCT_NAME,
  6  OD.QUANTITY,
  7  O.TOTAL_AMOUNT,
  8  PY.PAYMENT_METHOD,
  9  PY.PAYMENT_STATUS,
 10  PY.PAYMENT_DATE
 11  FROM CUSTOMER C
 12  INNER JOIN ORDERS O
 13  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 14  INNER JOIN ORDER_DETAILS OD
 15  ON O.ORDER_ID = OD.ORDER_ID
 16  INNER JOIN PRODUCT P
 17  ON OD.PRODUCT_ID = P.PRODUCT_ID
 18  LEFT JOIN PAYMENT PY
 19  ON O.ORDER_ID = PY.ORDER_ID
 20  ORDER BY C.CUSTOMER_ID, O.ORDER_DATE DESC;
```

```
CUSTOMER_ID CUSTOMER_NAME ORDER_ID ORDER_DAT PRODUCT_NAME          QUANTITY TOTAL_AMOUNT
----------- ------------- -------- --------- --------------------- -------- ------------
PAYMENT_METHOD   PAYMENT_STATUS PAYMENT_D
---------------- -------------- ---------
        101 Priya             204 23-SEP-26 Vitamin C Serum              1         1299
Cash on Delivery Pending        23-SEP-26

        101 Priya             201 10-SEP-26 Matte Lipstick               2         1797
UPI              Success        10-SEP-26

        101 Priya             201 10-SEP-26 Volume Mascara               1         1797
UPI              Success        10-SEP-26

        102 Ananya            205 25-SEP-26 Hydrating Moisturizer        1          899


        102 Ananya            202 11-SEP-26 Vitamin C Serum              1         1299
Credit Card      Success        11-SEP-26

        103 Kavya             203 12-SEP-26 Volume Mascara               1          799
Debit Card       Success        12-SEP-26
```

7. Customer Purchase History with Payment:

```
SQL> SELECT C.CUSTOMER_NAME,
  2  P.PRODUCT_NAME,
  3  SUM(OD.QUANTITY) AS TOTAL_QUANTITY,
  4  SUM(OD.QUANTITY * OD.PRICE) AS PRODUCT_SALES_AMOUNT
  5  FROM CUSTOMER C
  6  INNER JOIN ORDERS O
  7  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  8  INNER JOIN ORDER_DETAILS OD
  9  ON O.ORDER_ID = OD.ORDER_ID
 10  INNER JOIN PRODUCT P
 11  ON OD.PRODUCT_ID = P.PRODUCT_ID
 12  GROUP BY C.CUSTOMER_NAME, P.PRODUCT_NAME
 13  ORDER BY PRODUCT_SALES_AMOUNT DESC;
```

```
CUSTOMER_NAME PRODUCT_NAME          TOTAL_QUANTITY PRODUCT_SALES_AMOUNT
------------- --------------------- -------------- --------------------
Ananya        Vitamin C Serum                    1                 1299

Priya         Vitamin C Serum                    1                 1299

Priya         Matte Lipstick                     2                  998

Ananya        Hydrating Moisturizer              1                  899

Kavya         Volume Mascara                     1                  799

Priya         Volume Mascara                     1                  799
```

8. Multi-Table Business Report

The following query generates a complete customer purchase report.

```
SQL> SELECT PY.PAYMENT_ID,
  2  O.ORDER_ID,
  3  C.CUSTOMER_NAME,
  4  PY.PAYMENT_METHOD,
  5  PY.PAYMENT_DATE,
  6  PY.AMOUNT,
  7  PY.PAYMENT_STATUS
  8  FROM PAYMENT PY
  9  INNER JOIN ORDERS O
 10  ON PY.ORDER_ID = O.ORDER_ID
 11  INNER JOIN CUSTOMER C
 12  ON O.CUSTOMER_ID = C.CUSTOMER_ID
 13  ORDER BY PY.PAYMENT_DATE DESC;
```

```
PAYMENT_ID ORDER_ID CUSTOMER_NAME PAYMENT_METHOD   PAYMENT_D AMOUNT PAYMENT_STATUS
---------- -------- ------------- ---------------- --------- ------ --------------
       404      204 Priya         Cash on Delivery 23-SEP-26   1299 Pending

       403      203 Kavya         Debit Card       12-SEP-26    799 Success

       402      202 Ananya        Credit Card      11-SEP-26   1299 Success

       401      201 Priya         UPI              10-SEP-26   1797 Success
```

9. Payment Transaction Report:

```
SQL> SELECT C.CUSTOMER_NAME,
  2  O.ORDER_ID,
  3  O.ORDER_DATE,
  4  O.TOTAL_AMOUNT,
  5  PY.PAYMENT_METHOD,
  6  PY.AMOUNT AS PAYMENT_AMOUNT,
  7  PY.PAYMENT_STATUS
  8  FROM CUSTOMER C
  9  INNER JOIN ORDERS O
 10  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11  INNER JOIN PAYMENT PY
 12  ON O.ORDER_ID = PY.ORDER_ID
 13  ORDER BY O.ORDER_DATE DESC;
```

```
CUSTOMER_NAME ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_METHOD   PAYMENT_AMOUNT PAYMENT_STATUS
------------- -------- --------- ------------ ---------------- -------------- --------------
Priya              204 23-SEP-26         1299 Cash on Delivery           1299 Pending

Kavya              203 12-SEP-26          799 Debit Card                  799 Success

Ananya             202 11-SEP-26         1299 Credit Card                1299 Success

Priya              201 10-SEP-26         1797 UPI                        1797 Success
```